Year 12 Maths - Polynomials

Similar questions - Polynomial identities and substitution

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 In this question you must show detailed reasoning.

(a) Show that $(x-1)^4 - 5(x-1)^2 + 4 \equiv x^4 - 4x^3 + x^2 + 6x$. **[2]**

(b) Hence solve the equation $x^4 - 4x^3 + x^2 + 6x = 0$. **[4]**

2

(a) Factorise $x^4 - 10x^2 + 9$ into four linear factors. **[3]**

(b) Hence solve the inequality $x^4 - 10x^2 + 9 \leq 0$, giving your answer in set notation. **[3]**

Question 1

(a) $(x-1)^4 = x^4 - 4x^3 + 6x^2 - 4x + 1$ and $-5(x-1)^2 = -5x^2 + 10x - 5$. Adding 4: $x^4 - 4x^3 + x^2 + 6x + 0$

(b) Let $u = (x-1)^2$: $u^2 - 5u + 4 = 0 \Rightarrow (u-1)(u-4) = 0$, so $(x-1)^2 = 1$ or 4. Hence $x - 1 = \pm 1$ or ± 2, giving $x = 2, 0, 3, -1$

Question 2

(a) With $u = x^2$: $u^2 - 10u + 9 = (u-1)(u-9)$, so $x^4 - 10x^2 + 9 = (x-1)(x+1)(x-3)(x+3)$

(b) A positive quartic with roots $-3, -1, 1, 3$ is negative between the 1st and 2nd roots and between the 3rd and 4th: $\{x : -3 \leq x \leq -1\} \cup \{x : 1 \leq x \leq 3\}$